

LOOK UP: gradient descent, QLearning

LIMIT THEOREMS

\* Pointwise Convergence

We say that  $\{x_n\}$  converges to  $x \in \mathbb{R}$  (denoted by  $x_n \rightarrow x$ ) if for every  $\epsilon > 0$ , we can find an  $N(\epsilon) \in \mathbb{N}$  such that  $|x_n - x| < \epsilon$  for  $n > N(\epsilon)$ .

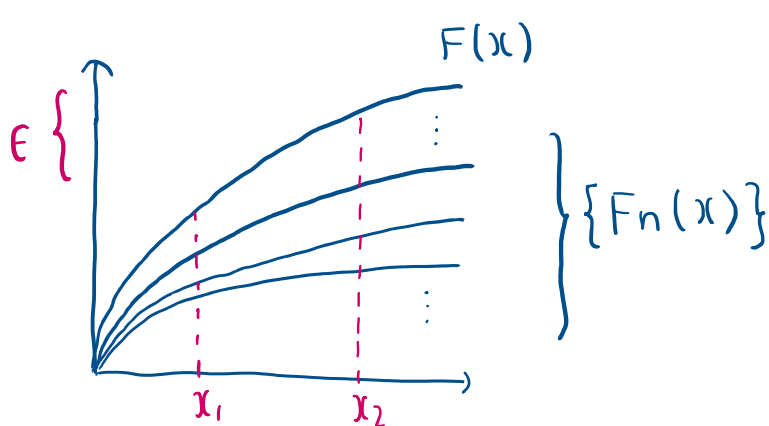
We say that the sequence of functions  $F_n(\cdot)$  converges to  $F(\cdot)$  pointwise if the sequence  $\{F_n(x)\}$  converges to  $F(x)$  ( $F_n(x) \rightarrow F(x)$ ) for all  $x \in \mathbb{R}$ .

\* Uniform Convergence

For every  $x$ , the sequence  $\{F_n(x)\}$  converges to  $F(x)$ .

For every  $\epsilon$ , there exists  $N(\epsilon, x)$  which can depend on  $x$ .

Only those  $F_n(x)$  for which  $n > N(\epsilon, x)$  are  $\epsilon$  close to  $F(x)$ .



$\therefore$  If  $N(\epsilon, x) = N(\epsilon)$ , i.e. independent of  $x$ , for every  $x \in \mathbb{R}$ , then such convergence of  $F_n(\cdot)$  to  $F(\cdot)$  is called as uniform convergence.

\* Convergence of Sequence of random variables

We are now interested in the convergence properties of an infinite sequence of random variables  $\{X_n\}$  to some limiting random variable  $X$ .

When you perform the random experiment once, you get a sequence of realisations  $\{X_n\}$  and  $x$ . could be a deterministic number

① POINTWISE OR SURE CONVERGENCE

$\{X_n, n \geq 0\}$  converges to  $X$  pointwise or surely if for all  $\omega \in \Omega$ , we have  $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$

e.g. Consider  $\Omega = \{H, T\}$

$$X_n = \begin{cases} 1/n & \text{if } \omega = H \\ 1 + 1/n & \text{if } \omega = T \end{cases} \quad \text{and } X = \begin{cases} 0 & \text{if } \omega = H \\ 1 & \text{if } \omega = T \end{cases}$$

② ALMOST SURE CONVERGENCE

$X_n$  converges to  $X$  almost surely if  $P(\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)) = 1$

The set of outcomes where the convergence does not happen has measure 0.

i.e.  $P(\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)) = 0$

e.g.  $\Omega = [0, 1]$   
You pick a number uniformly in  $[0, 1]$ .

$$X_n = \omega^n \quad \text{for all } \omega \in \Omega$$
$$X(\omega) = 0 \quad \text{for all } \omega$$

$$X_n(\omega) \rightarrow X(\omega) \quad \text{for } \omega \in [0, 1)$$
$$X_n(\omega) \not\rightarrow X(\omega) \quad \text{for } \omega = 1$$

$$P(\{1\}) = 0 \quad \text{and } P([0, 1)) = 1 \quad \therefore \text{almost sure a.s. convergence}$$

e.g. Strong law of large numbers (SLLN)

Let  $\{X_n, n \geq 0\}$  denote a sequence of i.i.d random variables with mean  $\mu$  and  $S_n = \sum_{i=1}^n X_i$ . Then,  $\frac{S_n}{n} \rightarrow \mu$  a.s.

Incremental formula for sample mean

We know  $\frac{S_n}{n} \rightarrow \mu$ .

We can use  $\hat{\mu}_n = \frac{S_n}{n}$  as an estimator for the mean

especially in cases where the underlying distribution is not known.

Note that  $\hat{\mu}_n$  is a random variable.

$$E[\hat{\mu}_n] = \mu \Rightarrow \hat{\mu}_n \text{ is an unbiased estimator.}$$

$$\text{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}$$

Now given  $\hat{\mu}_n$ , suppose you see an additional sample  $X_{n+1}$ .

How will you compute  $\hat{\mu}_{n+1}$ ?

① Naive way:  $\hat{\mu}_{n+1} = \frac{\sum_{i=1}^{n+1} X_i}{n+1}$

② Incremental formula

$$\hat{\mu}_n = \frac{S_n}{n} \Rightarrow S_n = \hat{\mu}_n \cdot n$$

$$S_{n+1} = S_n + X_{n+1} \Rightarrow \hat{\mu}_{n+1} = \frac{S_{n+1}}{n+1} = \frac{S_n + X_{n+1}}{n+1} = \frac{\hat{\mu}_n \cdot n + X_{n+1}}{n+1}$$

$$\therefore \hat{\mu}_{n+1} = \hat{\mu}_n + \frac{1}{n+1} [X_{n+1} - \hat{\mu}_n]$$